CMSC730 - Spring 2026 [Assignment 3]
University of Maryland

Light Game

Abstract

In this assignment, you will design and implement a simple interactive "light game" using the ESP32 microcontroller and its connected peripherals. The purpose of this project is to strengthen your understanding of sensor input, real-time feedback, and user interaction in embedded systems. By combining elements such as light sensing, sound generation, and user control, you will create a playful and engaging system that responds dynamically to environmental changes. This exercise encourages creativity while reinforcing core programming concepts such as input processing, signal mapping, and timing control.

Description

The light game will require your program to generate and interpret sensor data while providing continuous feedback to the player. The gameplay will involve adjusting the ambient light level to match a target value chosen at random by the system. The player will use physical actions, such as shading or exposing the light sensor, to bring the measured light intensity as close as possible to the generated target. Sound feedback from the buzzer will guide the player throughout the game.

Below are the basic requirements for implementing the light game:

1. At the beginning of the game, the system should generate a random number that represents the target light intensity to be reached by the player.
2. The target number corresponds to a specific level of ambient light detected by the light sensor connected to the ESP32.
3. When the game begins, play a short melody through the buzzer to signal that the system is ready and that the player can start.
4. Once the game starts, the player can change the lighting conditions around the sensor by increasing or decreasing the amount of light until the sensor reading approaches the target value.
5. The buzzer should emit a continuous tone that changes in pitch according to how close the player is to the target light intensity.
6. As the player moves closer to the target value, the pitch of the buzzer should rise gradually, giving clear auditory feedback that they are improving.
7. When the player moves further away from the target, the pitch of the buzzer should drop, indicating that the current light level is less accurate.
8. If the player successfully reaches the target light intensity within ten seconds, the buzzer should play a short winning melody to signal success.
9. If the player fails to reach the target within the given time, a different melody should be played to indicate that the game is over.

10. After the game ends, the player can press a designated key on the keyboard to restart the game and begin a new round with a freshly generated target value.

Your implementation should clearly demonstrate the use of both hardware and software elements of the ESP32 system. You are expected to write efficient code that continuously reads the sensor values, maps them to corresponding buzzer frequencies, and evaluates whether the player has reached the target. Timing control should be handled precisely so that the ten-second limit and sound feedback remain consistent.

Be sure to test your design under different lighting conditions to verify that the game behaves predictably and that the buzzer feedback remains responsive. By the end of this assignment, you should have a functioning light-based game that effectively integrates sensors, actuators, and user interaction through sound and light, illustrating your ability to design a small but complete interactive system.

Delivery

There are two deliverables.

- Upload your code. You can put them in a zip file if needed.
- a quick video demo of the working game with an (unlisted) youtube video link.

Due Date

Th Mar 5th, 11:59 PM EST

Last Modified Date: Feb 26th, 2026